

Early Identification of Heart Attack vs Gastritis Using AI/ML

Project ID: R26-IT-012 | Year: 2026



Our Supervisors

Supervisor



Mr. SMB Harshanath
Senior Lecturer

Co-supervisor



Prof. Samantha Rajapaksha
Head | Department of
Information Technology

External
Supervisor



Dr. Nishan Vas
Consultant Cardiologist
TH Polonnaruwa

External
Supervisor



Dr S.M.K.G.H.Samarakoon
Senior Medical officer
Cardiology TH Polonnaruwa

Team Members



Student Name	Student ID
Jayalath J.R.D.S	IT22222640
Peiris B.A	IT22193490
Wijewardhana I.L	IT22116024
Maheesha S.L	IT22609380

INTRODUCTION

Real-World Problem

- Chest pain is a common symptom in emergency departments and clinical training environments.
- Gastric diseases such as gastritis and GERD can produce symptoms similar to heart attacks, making early diagnosis difficult.
- Medical students often struggle to differentiate cardiac and gastric chest pain during clinical learning.
- Traditional learning methods rely mainly on textbooks and limited patient exposure.
- There is a need for an AI-assisted educational system that helps medical students understand the diagnostic workflow from symptoms to treatment decisions.

Scenario Demonstration(2-Minute Video)



Users and Stakeholders

Primary Users

- Medical Students

Secondary Users

- Doctors and Clinical Practitioners
- Medical Educators / Universities

Stakeholders

- Hospitals and Healthcare Institutions
- Medical AI Researchers and Developers
- Healthcare Systems / Patients

Existing Solutions and Their Limitations

Existing Solutions

- AI systems for cardiac disease detection using ECG signals.
- AI models for gastrointestinal disease detection using **endoscopy** images.
- Clinical scoring methods used in emergency triage.
- Medical education tools mainly based on theoretical case studies.

Limitations of Current Systems

- Most systems analyze only one type of medical data.
- Limited integration of symptoms, diagnostic tests, and medical imaging.
- Existing systems focus mainly on clinical diagnosis rather than student learning.
- Lack of systems showing step-by-step clinical reasoning from symptoms to treatment.

Proposed Solution

The proposed system integrates patient symptoms, cardiac data, and medical imaging to help medical students understand how doctors differentiate cardiac and gastric chest pain.

System Components

1. Baseline Risk & Patient Profile Analysis
 - Analyze symptoms and risk factors using machine learning.
2. ECG & Echocardiography Analysis Module
 - ECG and Troponin analysis to detect cardiac abnormalities.
3. Angiogram Evaluation & Treatment Recommendation
 - AI analysis of coronary angiography images.
4. Gastric Analysis and Decision Support Engine
 - Deep learning analysis of endoscopy images.

Functional Requirements

- The system shall **receive symptom data and gastroscopy images** for analysis.
- The system shall **analyze gastroscopy images using an AI model** to detect gastric diseases.
- The system shall **classify gastric disease severity** (Low, Medium, High).
- The system shall **generate treatment recommendations and display results**.

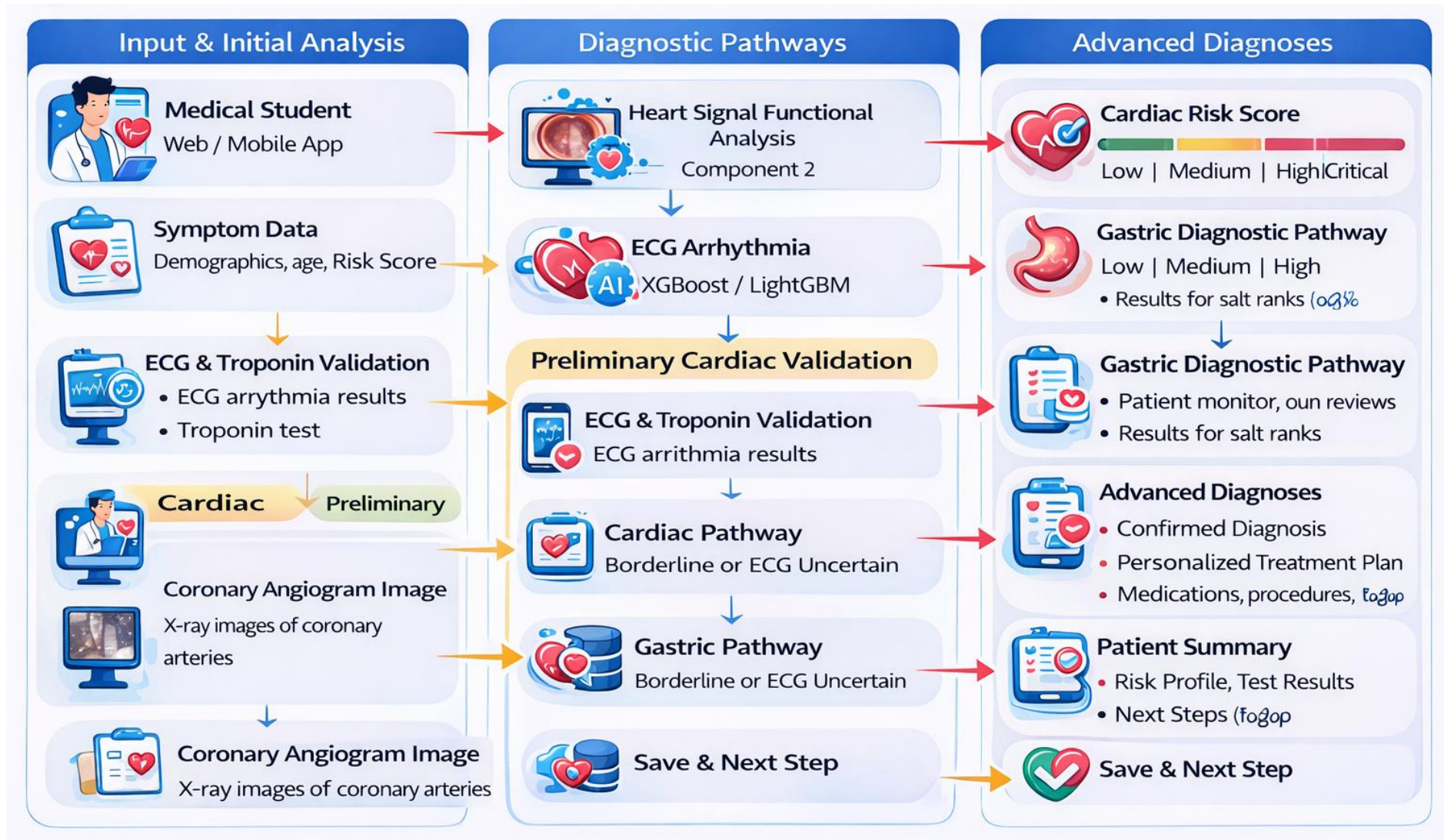
Non-Functional Requirements

- The system shall **provide accurate and reliable AI predictions**.
- The system shall **ensure patient data privacy and security**.
- The system shall **process medical images efficiently with acceptable response time**.
- The system shall **provide a user-friendly interface for medical students and clinicians**.

BUDGET & JUSTIFICATION

Item	Component / Description	Estimated Cost (LKR)	Justification
1. Dataset Collection & Preparation	Medical datasets for ECG, angiogram, and gastroscopy images; annotation and preprocessing	5,000	Required for training AI models to detect cardiac and gastric conditions accurately.
2. AI Model Development	Training deep learning models (CNN, EfficientNet, U-Net) using GPU/cloud resources	8,000	Enables automated detection of artery blockages, ECG abnormalities, and gastric diseases.
3. System Development	Web dashboard, backend APIs, and AI model integration using Python, FastAPI, and React	4,500	Needed to build the user interface and integrate all system components for clinical use.
4. Infrastructure & Storage	Cloud hosting, database (PostgreSQL), and storage for medical images and AI results	3,000	Supports secure storage and real-time processing of medical imaging data.
TOTAL ESTIMATED BUDGET		20,500 LKR	

System Overview Diagram



How the Proposed Solution Overcomes Limitations

- Integrates multi-modal clinical data (symptoms, ECG, medical images).
- Demonstrates a complete diagnostic workflow from screening to treatment decision.
- Provides AI-assisted cardiac and gastric disease detection.
- Allows medical students to suggest prescriptions and compare with AI recommendations.
- Supports clinical reasoning and case-based medical learning.

IT22193490 | PEIRIS.B.A

Information Technology

Baseline Risk & Patient Profile Analysis



Contribution to Overall Solution:

- Performs **early patient screening** using questionnaire-based clinical data
- Calculates **dual risk scores** to estimate cardiac and gastric probability
- Identifies patients requiring **urgent cardiac investigation**
- **Generates a preliminary diagnostic category** to guide the next diagnostic pathway

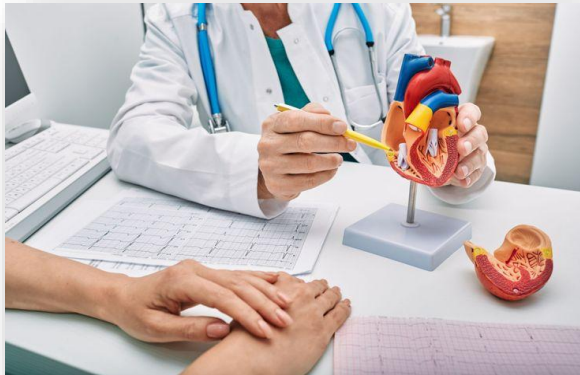
Novelty:

- **Dual-risk scoring mechanism** to differentiate cardiac and gastric chest pain
- **Hybrid approach** combining rule-based clinical scoring with machine learning
- Integration of **symptoms, lifestyle, and medical history factors** for early screening
- **Privacy-preserving design** using anonymized clinical attributes

Baseline Risk & Patient Profile Analysis

Problem Definition :

- Chest pain caused by **heart attacks and gastric diseases often shows similar symptoms**
- **Many patients misinterpret cardiac chest pain as gastritis**, delaying medical treatment
- Existing AI diagnostic systems **mainly focus only on cardiac disease prediction**
- **There is a need for an early screening system that differentiates cardiac and gastric causes of chest pain**



Research Objective:

Main Objective:

- Develop an **AI-assisted baseline risk assessment system** that analyzes patient symptoms, medical history, and lifestyle factors to differentiate **cardiac and gastric chest pain** during early patient screening.

Sub –Objectives:

- Collect and analyze **patient demographic, lifestyle, and symptom data**
- **Implement a dual-risk scoring model** for cardiac and gastric probability
- Apply **machine learning techniques (XGBoost / LightGBM)** for risk prediction
- Support **clinical reasoning for medical students through AI decision support**

Baseline Risk & Patient Profile Analysis

Research Methodology (Design Science Research)

- Uses **Design Science Research (DSR)** methodology to design and evaluate an AI-assisted diagnostic component.
- Collects **patient questionnaire data**, including demographic information, medical history, lifestyle factors, and symptom characteristics.
- Applies a **dual-risk scoring mechanism** to calculate both cardiac and gastric risk probabilities.
- Combines **rule-based clinical scoring with machine learning models** to improve early risk prediction.
- Generates a **preliminary diagnostic category** that determines the next diagnostic pathway.

Technologies Used

-  **Python** – primary programming language for machine learning and backend development
-  **FastAPI** – backend framework for integrating AI models with the web interface
-  **Scikit-learn, XGBoost, LightGBM** – machine learning algorithms used for structured clinical data analysis
-  **Pandas & NumPy** – data preprocessing and feature engineering
-  **PostgreSQL** – database for storing questionnaire data and prediction results
-  **SHAP (Explainable AI)** – used to interpret model predictions and highlight key risk factors

Baseline Risk & Patient Profile Analysis

Validation and Evaluation

- **Model performance evaluation** using metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC
- **Training and testing datasets** used to evaluate the reliability of the machine learning models
- **Cross-validation techniques** applied to ensure model generalization across different patient profiles
- **Clinical scenario simulation** used to test system predictions using different symptom combinations
- Evaluation ensures the system can **accurately identify cardiac risk during early patient screening**

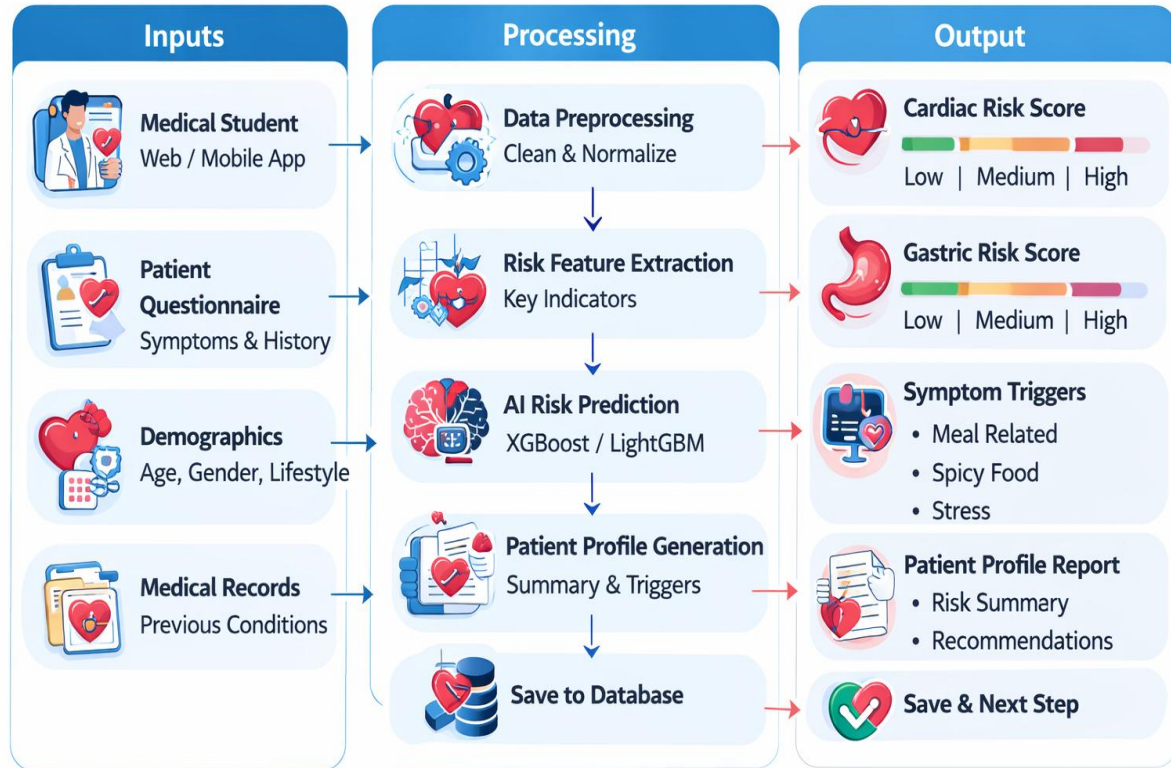
Data, Security and Ethical Considerations

- **No personally identifiable information (PII)** such as patient names or contact details is collected
- System uses **anonymized clinical attributes** including age, symptoms, lifestyle factors, and medical history
- Training datasets are obtained from **public medical datasets and simulated patient records**
- **Secure database storage (PostgreSQL)** used for storing questionnaire responses and prediction results

Baseline Risk & Patient Profile Analysis

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

Component 2 – Heart Signal Analysis

High cardiac-risk patients are forwarded for ECG and echocardiography analysis.

Component 3 – Angiogram Evaluation

Confirmed cardiac cases proceed to angiogram analysis and treatment recommendation.

Component 4 – Gastric Analysis

If cardiac risk is low, the case is redirected to gastric disease detection.

Shared Clinical Database

All components share patient data, risk scores, and diagnostic results.

IT22609380 | MAHEESHA S.L

Information Technology

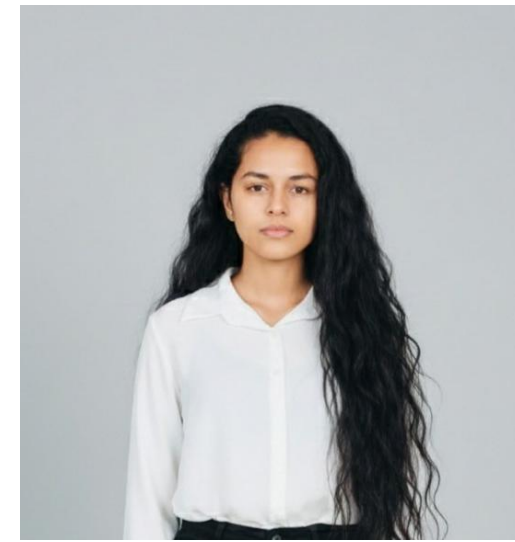
Heart Signal & Functional Analysis

Contribution to Overall Solution:

- Detects cardiac abnormalities using **ECG and Echo analysis**.
- Helps differentiate **cardiac vs gastric chest pain**.
- Provides **AI-based diagnostic support**.
- Supports **medical student learning and decision making**.

Novelty:

- Combines **ECG and Echo image analysis**.
- Uses **deep learning for cardiac abnormality detection**.
- Designed as an **AI clinical decision support tool**.
- Enables **early cardiac screening before gastric analysis**.



Heart Signal & Functional Analysis

Problem Definition :

- Chest pain causes **cardiac or gastric confusion**.
- **Heart diseases can be misdiagnosed** as gastric issues.
- ECG and Echo **require expert interpretation**.
- Need **AI support for early cardiac detection**.



Research Objective:

Main Objective:

- Develop **AI-based ECG & Echo analysis for cardiac abnormality detection**.

Sub – Objectives:

- Detect **abnormal ECG patterns** using AI.
- Analyze **echocardiogram images**.
- Combine ECG + Echo **for cardiac assessment**.
- Provide **structured diagnostic outputs that support the overall AI diagnostic system**.

Heart Signal & Functional Analysis

Research Methodology (Design Science Research)

- Identify the challenge of **differentiating cardiac chest pain** using ECG and echocardiogram data.
- Design an **AI-based model to analyze** ECG images and cardiac imaging data.
- Implement **deep learning algorithms for ECG image classification and echocardiogram interpretation.**
- Validate the model using **medical datasets and evaluated using metrics** such as accuracy, precision, and recall.

Technologies Used

-  **Python** – AI model development
-  **FastAPI** – Backend API services
-  **Pandas & NumPy** – Data processing
-  **PyTorch** – Training CNN models
-  **Matplotlib** – Visualization of ECG analysis results
-  **OpenCV** – ECG & Echo image processing
-  **PostgreSQL** – Storing prediction results

Heart Signal & Functional Analysis

Validation and Evaluation

- Evaluate model performance using **accuracy, precision, recall, and F1-score metrics.**
- Validate ECG classification using labeled datasets with **normal and abnormal patterns.**
- Assess echocardiogram accuracy by **testing image classification results on medical datasets.**
- Perform **system-level testing to ensure proper integration with other diagnostic components.**

Data, Security and Ethical Considerations

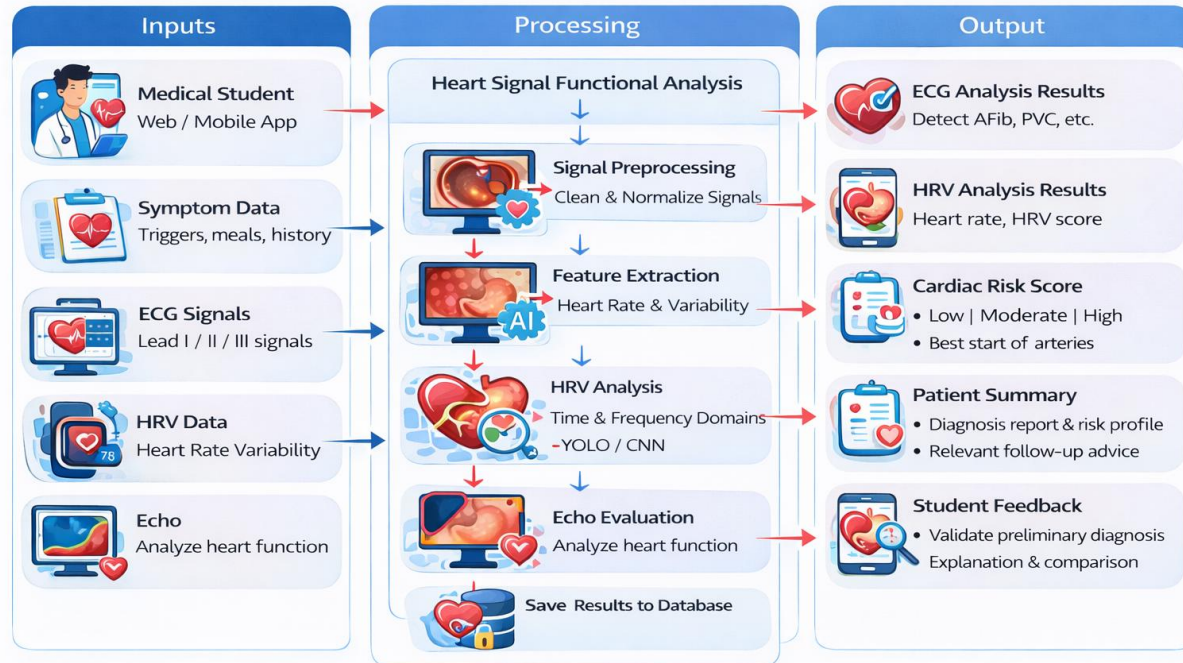
- Use **anonymized medical datasets.**
- Ensure **patient data privacy.**
- Apply **secure data storage.**
- Use system as **decision-support only.**



Heart Signal & Functional Analysis

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

- **Initial Patient Assessment** - Sends cases with cardiac risk for ECG and Echo analysis.
- **Angiogram Evaluation** - If cardiac abnormalities are detected, the case is forwarded to Component 3.
- **Gastric Analysis** - If cardiac abnormalities are not detected, the case is redirected to Component 4.
- **Shared Clinical Database** - Stores ECG images, Echo data, risk scores, and diagnostic results shared across all components.

IT22222640 | JAYALATH J.R.D.S

Information Technology



Gastric Analysis and Decision Support Engine

Contribution to Overall Solution:

- Diagnose **gastric causes of chest pain** after cardiac conditions are excluded.
- Analyze **symptom triggers and gastroscopy images** using AI models.
- Identify diseases such as **gastritis, GERD, and peptic ulcer**.
- Generate **AI-based treatment recommendations**.

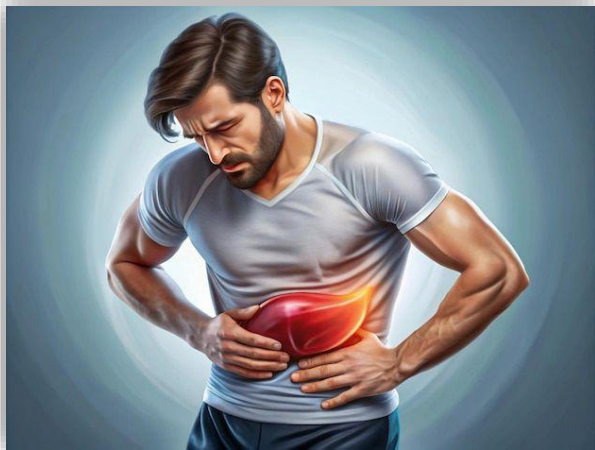
Novelty:

- Integration of **symptom analysis with AI image diagnosis**.
- **EfficientNet-B4 deep learning model** for gastroscopy image classification.
- **Severity classification** of gastric diseases.
- **AI-assisted prescription recommendation** for treatment.

Gastric Analysis and Decision Support Engine

Problem Definition :

- Gastric diseases can produce **chest pain similar to heart attacks**.
- Symptom-based diagnosis alone can lead to **misdiagnosis**.
- Accurate diagnosis requires **gastroscopy image analysis**.
- Need an **AI-assisted system for gastric disease detection**.



Research Objective:

Main Objective:

- Develop an **AI-based gastric analysis system** that detects gastric diseases and generates treatment recommendations.

Sub – Objectives

- Identify **gastric symptom triggers** from patient questionnaires.
- Train **deep learning model for gastroscopy image classification**.
- Combine **symptom analysis and image-based diagnosis**.
- Generate **AI-based treatment recommendations**.

Gastric Analysis and Decision Support Engine

Research Methodology(Design Science Research)

- Use **Design Science Research (DSR)** methodology.
- Develop an **AI-based gastric diagnosis system**.
- Apply **deep learning for gastroscopy image analysis**.
- Evaluate system performance using **classification metrics**.

Technologies Used

-  **Python, FastAPI** – backend and API services
-  **PyTorch, EfficientNet-B4** – deep learning model
-  **OpenCV, Albumentations** – image preprocessing
-  **React.js, PostgreSQL** – dashboard and database

Gastric Analysis and Decision Support Engine

Validation and Evaluation

- Dataset split into **training, validation, and testing sets**.
- Evaluate model using **accuracy, precision, recall, and F1-score**.
- Use **confusion matrix and ROC analysis** for performance evaluation.
- Verify predictions using **gastroscopy image analysis**.

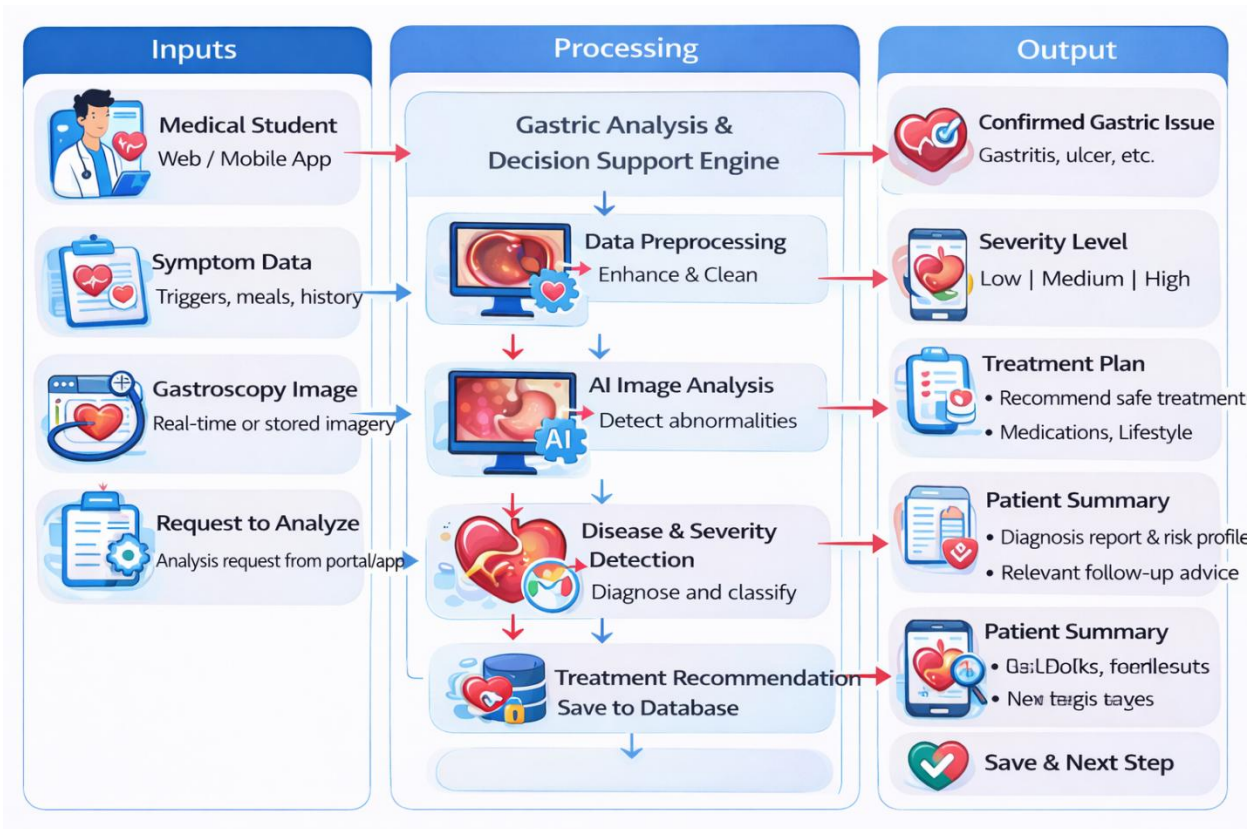
Data, Security and Ethical Considerations

- Use **anonymized medical datasets** for model training.
- Avoid collecting **personal patient identifiers**.
- Follow **ethical guidelines for medical AI research**.
- System designed to **assist doctors, not replace them**.

Gastric Analysis and Decision Support Engine

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

- **Patient symptom data** from Component 1 (questionnaire analysis and gastric risk prediction).
- **ECG and Troponin test** results to confirm that cardiac causes are excluded before gastric analysis.
- **Gastroscopy images** used by the AI model for gastric disease detection.
- **Preliminary gastric prediction** generated from earlier system modules.

IT22116024 | WIJEWARDHANA I.L

Information Technology

Angiogram Evaluation & Treatment Recommendation



Contribution to Overall Solution:

- **Analyzes angiogram images** and reports to detect artery blockages.
- **Identifies affected arteries** (LAD, RCA, LCX) and estimates stenosis percentage.
- **Classifies blockage severity** for accurate diagnosis.
- Generates **AI-assisted treatment recommendations** (medication, PCI, or CABG).

Novelty:

- AI-based angiogram analysis for **artery segmentation** and **blockage detection**.
- Explainable **AI visualization** highlighting **blockage** regions.
- Automated **treatment recommendations** based on **blockage severity**.
- **Patient-friendly visualizations** of coronary artery conditions.

Angiogram Evaluation & Treatment Recommendation

Problem Definition :

- **Coronary artery blockages** are a major cause of heart attacks.
- **Angiogram interpretation** requires **expert cardiology knowledge**.
- **Manual analysis** can be **time-consuming and inconsistent**.
- Need an **AI-assisted system** for **angiogram analysis and treatment recommendation**.



Research Objective:

Main Objective:

- Develop an AI-assisted system to detect **coronary artery blockages**, **estimate stenosis severity**, and **recommend treatments**.

Sub – Objectives

- Prepare coronary angiogram image datasets.
- Apply **image preprocessing** and **artery segmentation**.
- Detect blockages using **deep learning models**.
- Estimate stenosis **percentage** and **classify severity** and generate **visual explanations**

Angiogram Evaluation & Treatment Recommendation

Research Methodology (Design Science Research)

- Uses DSR methodology.
- Develops an AI-based angiogram evaluation component.
- Inputs: coronary angiogram images and diagnostic reports.
- Applies computer vision and deep learning for blockage detection.
- Generates visual explanations and treatment recommendations.

Technologies Used

-  **Deep Learning Models CNN** – blockage detection and coronary artery segmentation.
-  **Python, FastAPI** – backend and API services
-  **PyTorch, EfficientNet-B4** – deep learning model
-  **OpenCV, Albumentations** – image preprocessing
-  **React.js, PostgreSQL** – dashboard and database

Angiogram Evaluation & Treatment Recommendation

Validation and Evaluation

- Evaluate **blockage detection** using **Accuracy, Precision, Recall, and F1-score**.
- Assess **artery segmentation** using **IoU and Dice Similarity Coefficient**.
- Evaluate **stenosis estimation** using **Mean Absolute Error**.
- Collect **clinical feedback** to assess **system usability and visual explanations**.

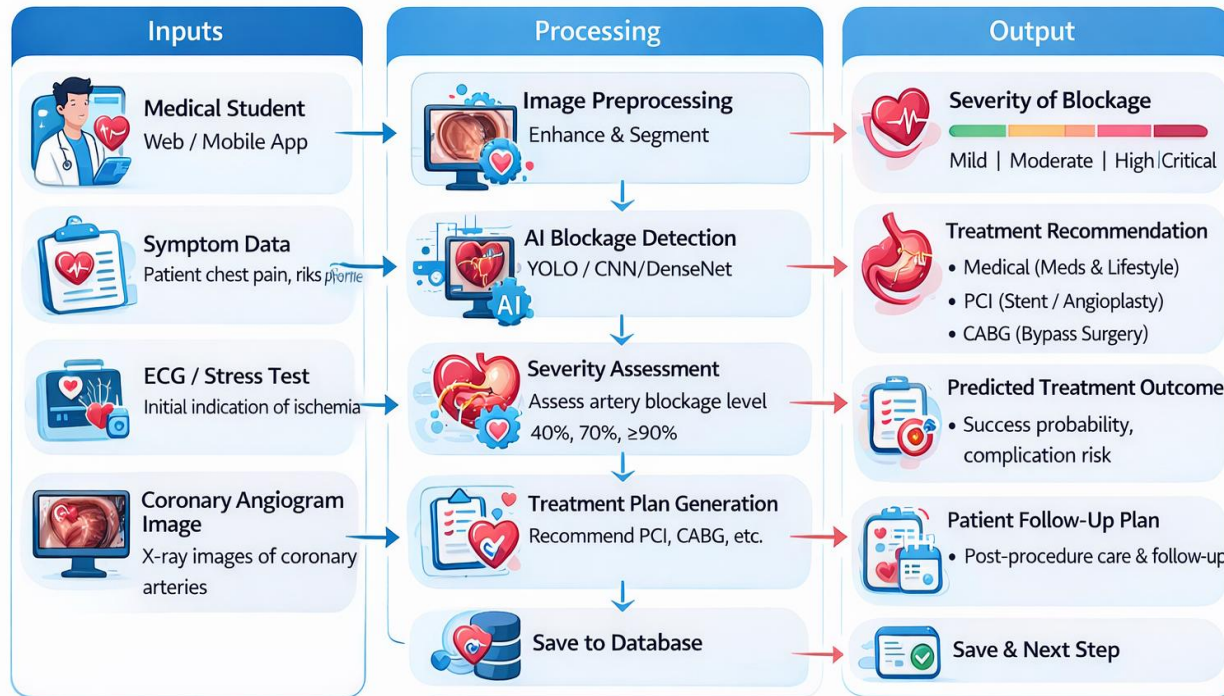
Data, Security and Ethical Considerations

- Use **anonymized angiogram datasets** from **medical repositories**.
- Ensure **patient privacy** and **secure data storage**.
- System acts as a **clinical decision support tool**, not a replacement for doctors.

Angiogram Evaluation & Treatment Recommendation

Implementation Details

System Overview of the Individual Component



Dependencies and Common Components

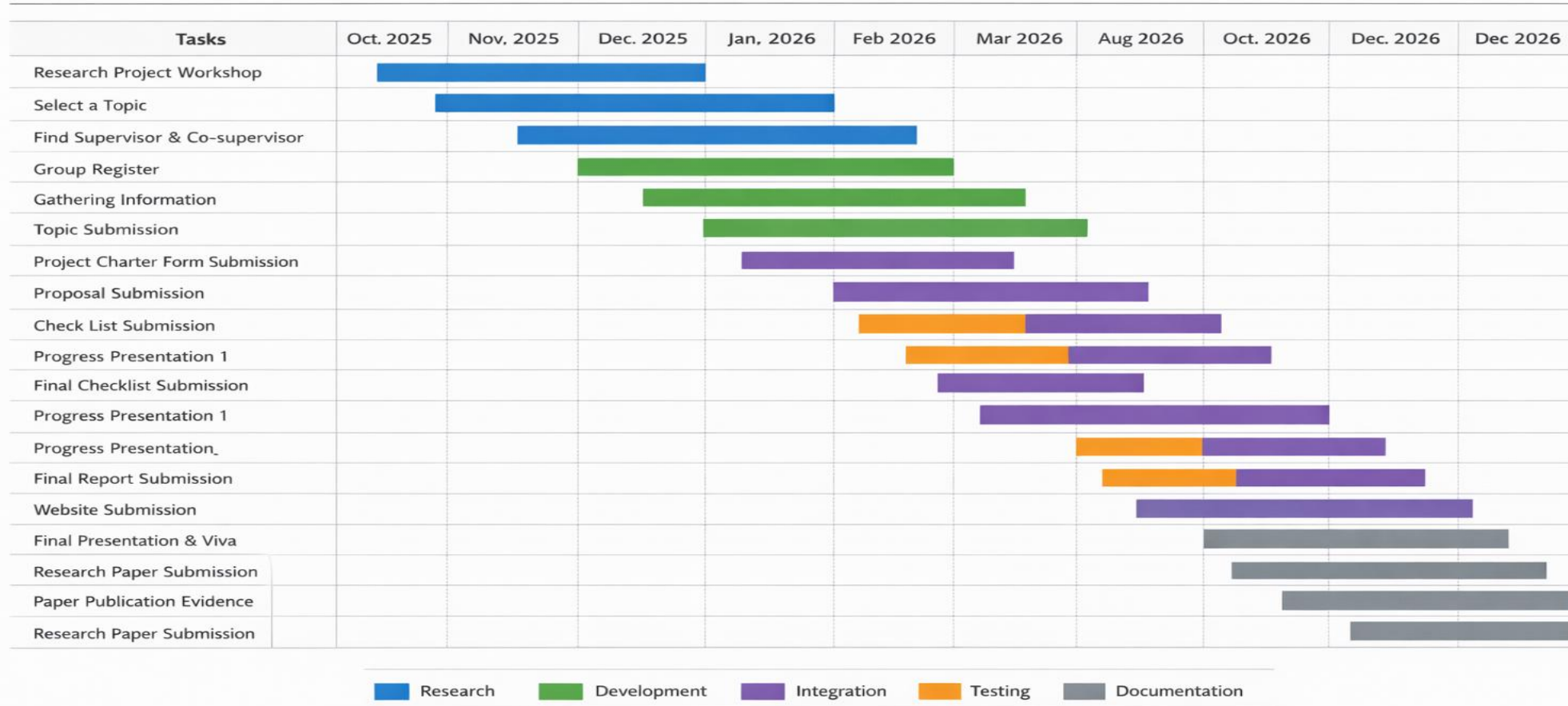
- **Computer Vision Models** – artery segmentation and blockage detection
- **Deep Learning Frameworks** – CNN / U-Net model processing
- **Backend APIs** – module communication and integration
- **Database / Cloud Storage** – image and result management
- **Shared Interface Components** – result display and user interaction

Business Model and Market Strategy

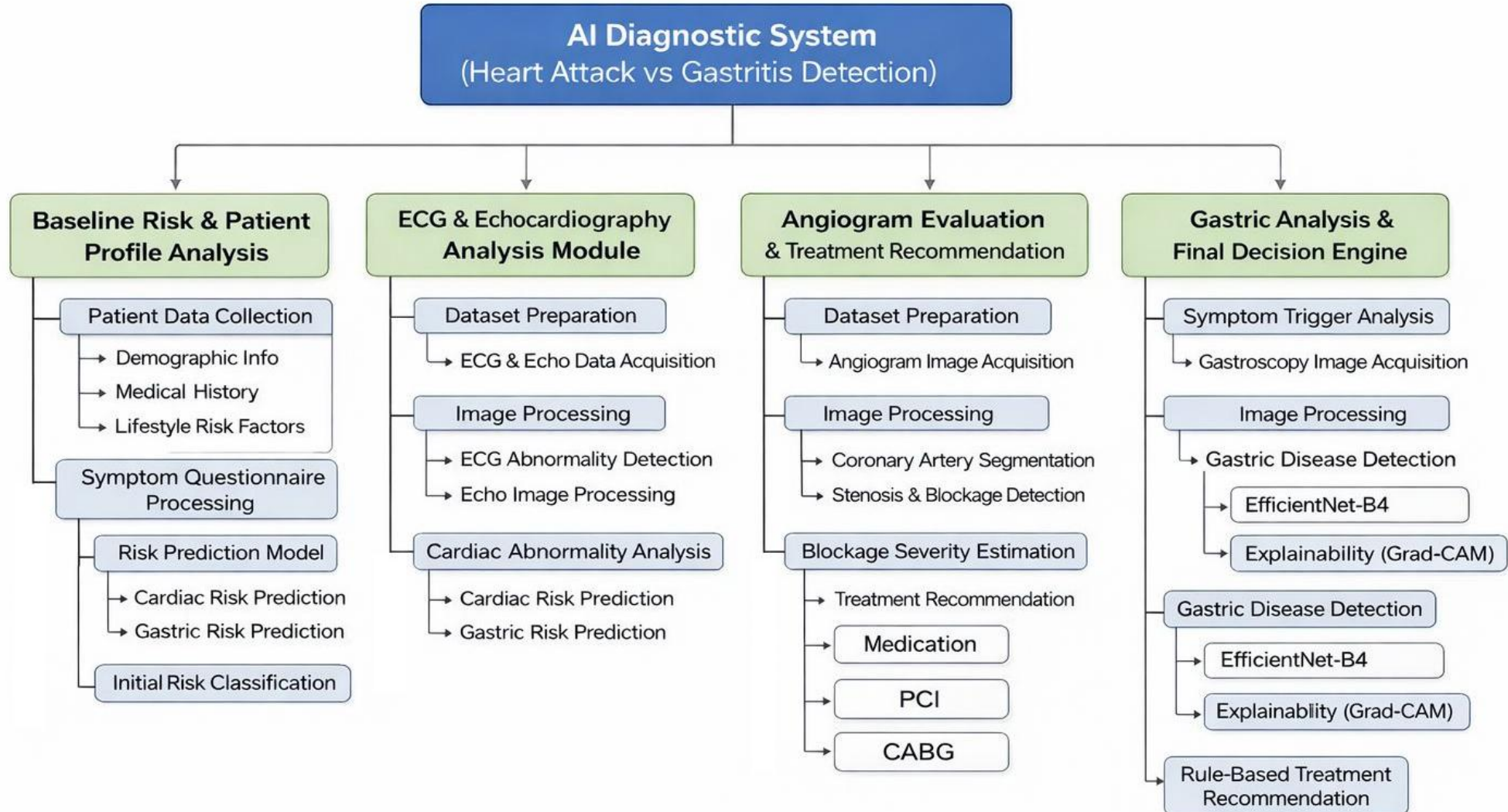
Target Market	Value Proposition	Investment / Cost	Revenue & Cost Recove	Intellectual Property (IP)
Primary Customers • Hospitals & Cardiac centers • Emergency departments • Diagnostic laboratorie	Early detection of cardiac vs gastric chest pain using AI Automated analysis of ECG, angiogram, and gastroscopy images AI-assisted treatment recommendation for clinicians Patient-friendly visual explanations of medical results	Development Costs • Medical dataset collection & annotation • AI model training (GPU / Cloud) • Web dashboard & backend development Infrastructure • Cloud servers & API hosting • Secure storage for medical images	Hospital Subscription Model • Monthly or annual hospital license Per-Analysis Service Fee • Fee for AI diagnostic reports Institutional Partnerships • Medical universities • Healthcare providers	AI diagnostic algorithms Integrated cardiac-gastric decision support system Angiogram visualization & treatment recommendation engine Software platform and medical AI models

Grantt Chart

Early Identification of Heart Attack vs Gastritis Using AI/ML Project Gantt Chart



WORK BREAKDOWN STRUCTURE



References

<https://arxiv.org/abs/1905.11946>

(EfficientNet – Deep learning model used for image classification)

<https://datasets.simula.no/hyper-kvasir/>

(HyperKvasir – Public dataset for gastrointestinal endoscopy images)

THANK YOU !